

ACFI315 Finance with Excel

Lecture 5 Mean-Variance Analysis



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Mean-Variance Framework

- Markowitz (1959) proposes the mean-variance framework.
- Mean and variance of returns can be treated as proxies of return and risk, respectively.
- If two assets yield the same return, investors will prefer the asset with less risk.
- If two assets have the same degree of risk, investors will prefer the asset with higher return.

Portfolio Theory

- Portfolio theory works out the 'best combination' of stocks to hold in your portfolio of risky assets.
- Allows investor to 'mix' or combine stocks to get the best return relative to the overall riskiness of the chosen portfolio.
- **Example:** What proportions of your own \$100 should you put in two different stocks (e.g. 'weights' = 25%, 75% which implies \$25, \$75).
- Different weights give rise to different risk-return combinations and this is the efficient frontier.
 - The choice of weight is influenced by the possibility of lending and borrowing.

Risk and Return Estimations

- For a portfolio of assets, the return is the weighted average of the returns from the individual assets, where the weight that is applied to each return, is the part of the total investment/fund.

$$R_{p,j} = \sum_{i=1}^N w_i R_{i,j}$$
$$R_p = E(R_p) = E\left(\sum_{i=1}^N w_i R_{i,j}\right)$$

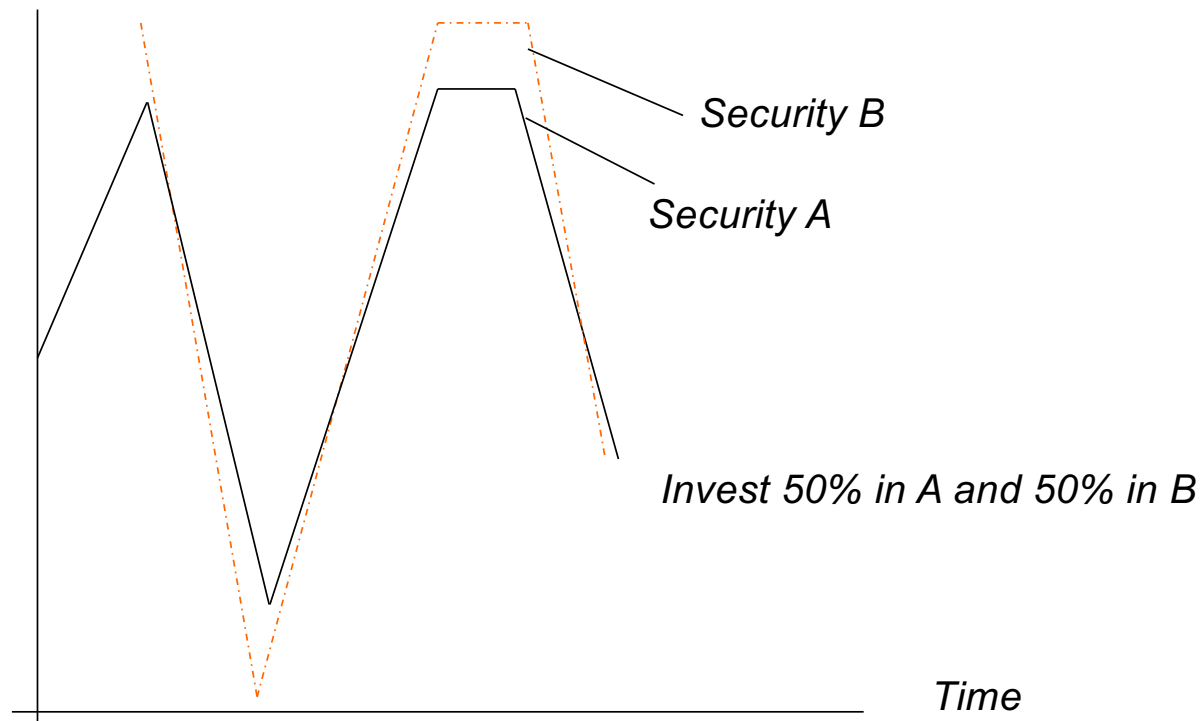
- The variance of a portfolio is the square of the deviation of the portfolio return from its mean and given by:

$$\sigma_p^2 = E\left[(R_p - \bar{R}_p)^2\right]$$

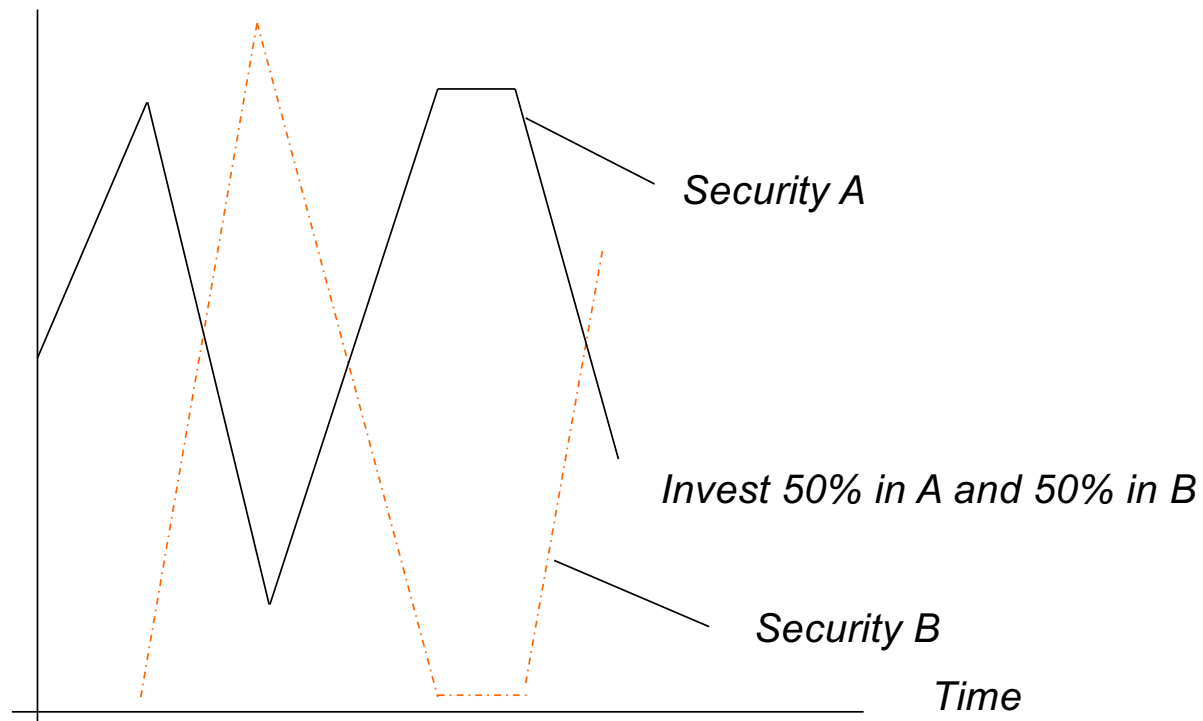
- For a portfolio with two assets:

$$\begin{aligned}\sigma_p^2 &= E\left[(R_p - \bar{R}_p)^2\right] = E\left[\left((w_1 R_{1,j} + w_2 R_{2,j}) - (w_1 \bar{R}_{1,j} + w_2 \bar{R}_{2,j})\right)^2\right] \\ &= E\left[\left(w_1 (R_{1,j} - \bar{R}_{1,j}) + w_2 (R_{2,j} - \bar{R}_{2,j})\right)^2\right] = w_1^2 \sigma_1^2 + w_2^2 \sigma_2^2 + 2w_1 w_2 \sigma_{12} \\ &= w_1^2 \sigma_1^2 + w_2^2 \sigma_2^2 + 2w_1 w_2 \sigma_1 \sigma_2 \rho_{12}\end{aligned}$$

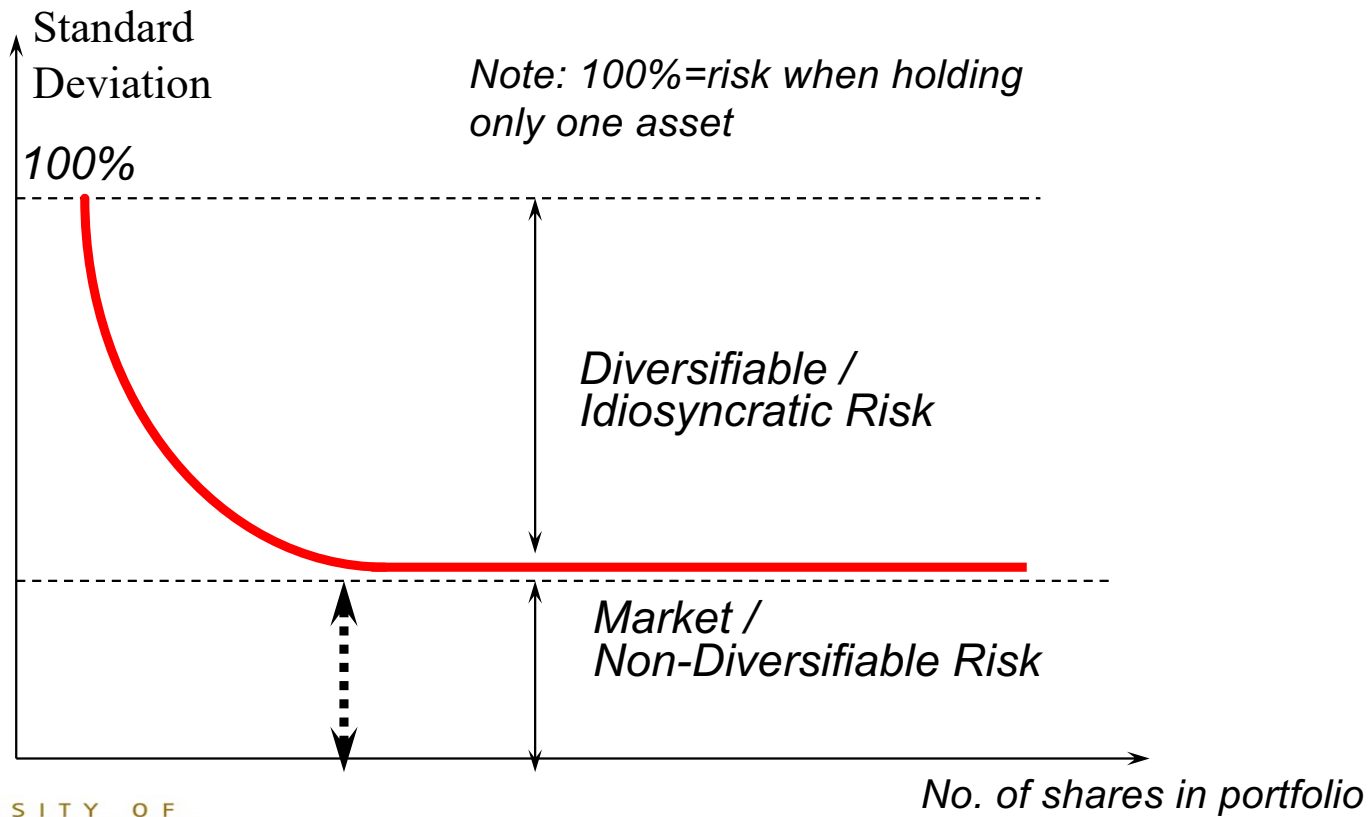
Perfect Positive Correlations



Perfect Negative Correlations

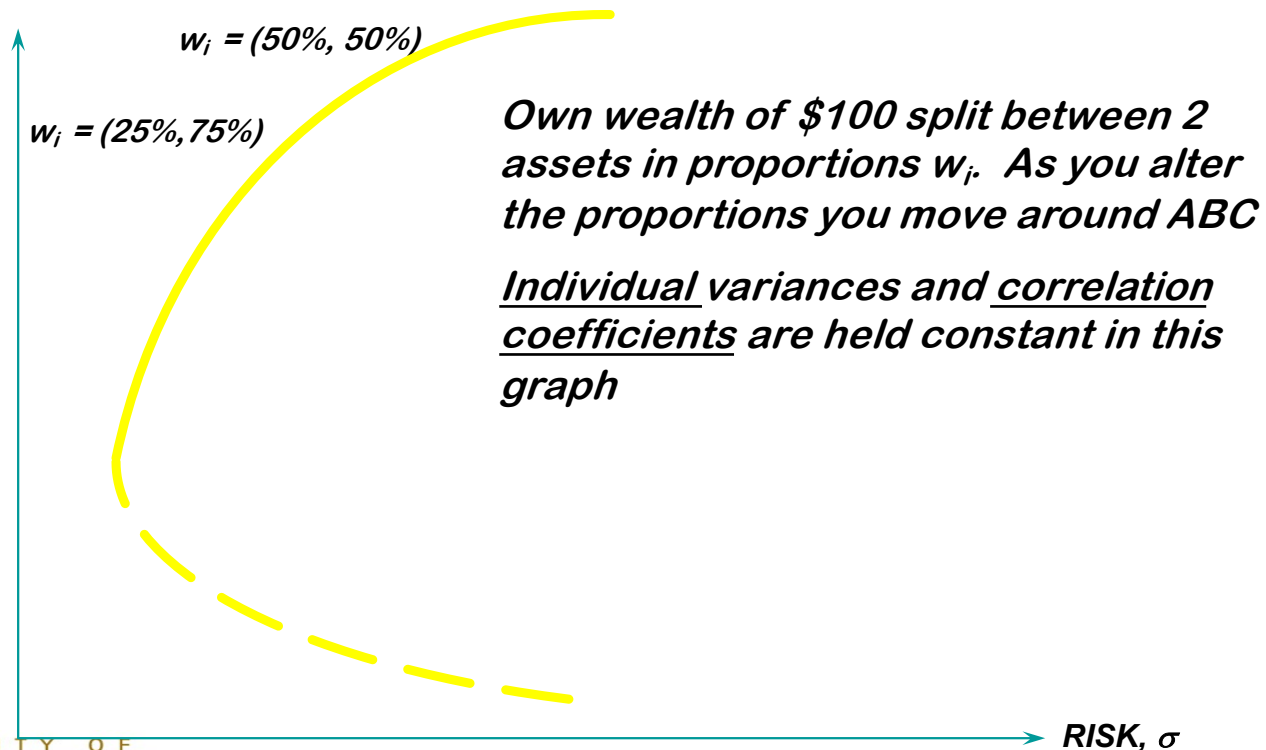


Diversification



Efficient Frontier

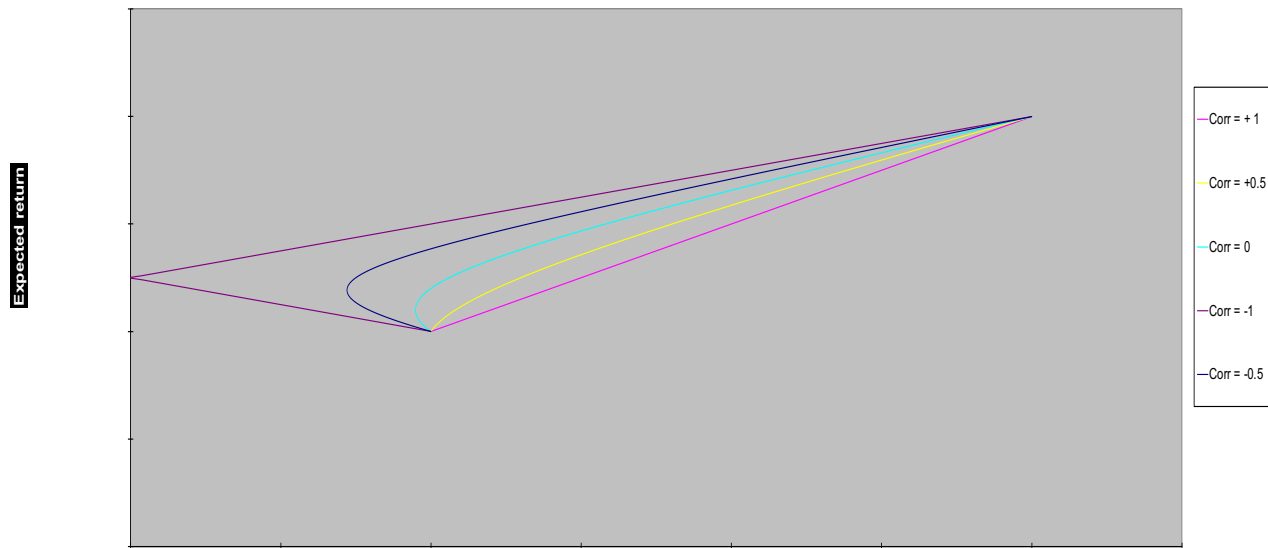
*Expected
Return*



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Impact of Correlations

Figure: Risk Reduction Through Diversification



Minimum Variance Portfolio

- The minimum variance portfolio is the portfolio consisting of two assets with the minimum variance.
- The weighting scheme for the minimum variance portfolio can be obtained by setting the first derivative of portfolio variance with respect to weight equal to 0.

$$\begin{aligned}\frac{\partial \sigma_p^2}{\partial w_1} &= \frac{\partial (w_1^2 \sigma_1^2 + w_2^2 \sigma_2^2 + 2w_1 w_2 \sigma_1 \sigma_2 \rho_{12})}{\partial w_1} = \frac{\partial (w_1^2 \sigma_1^2 + (1 - w_1)^2 \sigma_2^2 + 2w_1(1 - w_1) \sigma_1 \sigma_2 \rho_{12})}{\partial w_1} \\ &= \frac{\partial (w_1^2 \sigma_1^2 + \sigma_2^2 + w_1^2 \sigma_2^2 - 2w_1 \sigma_2^2 + 2w_1 \sigma_1 \sigma_2 \rho_{12} - 2w_1^2 \sigma_1 \sigma_2 \rho_{12})}{\partial w_1} \\ &= 2w_1 \sigma_1^2 + 2w_1 \sigma_2^2 - 2\sigma_2^2 + 2\sigma_1 \sigma_2 \rho_{12} - 4w_1 \sigma_1 \sigma_2 \rho_{12}\end{aligned}$$

- By setting $\frac{\partial \sigma_p^2}{\partial w_1} = 0$, we get:

$$\begin{aligned}2w_1 \sigma_1^2 + 2w_1 \sigma_2^2 - 2\sigma_2^2 + 2\sigma_1 \sigma_2 \rho_{12} - 4w_1 \sigma_1 \sigma_2 \rho_{12} &= 0 \\ 2w_1 \sigma_1^2 + 2w_1 \sigma_2^2 - 4w_1 \sigma_1 \sigma_2 \rho_{12} &= 2\sigma_2^2 - 2\sigma_1 \sigma_2 \rho_{12} \\ w_1 &= \frac{\sigma_2^2 - \sigma_1 \sigma_2 \rho_{12}}{\sigma_1^2 + \sigma_2^2 - 2\sigma_1 \sigma_2 \rho_{12}}\end{aligned}$$

Efficient Set Theorem – Risk Return Trade Off

- The efficient set theorem states that an investor will select his/her optimal portfolio from a set of possible portfolios which offers the:
 - Maximum expected return for every given level of risk.
 - Minimum risk for every given level of expected return.